MIL-G-29170A(YD)
17 December 1982
SUPERSEDING
MIL-G-29170(YD)
25 April 1977

MILITARY SPECIFICATION

GASKETS, SYNTHETIC RUBBER, REINFORCED (FOR DRYDOCK CAISSONS)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers molded, reinforced, medium-soft, oil-resistant, synthetic rubber gaskets for drydock caissons.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Std. 191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-190 - Identification Marking of Rubber Products.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5330

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D395 - Standard Test Methods for Rubber Properties - Compression Set.
D412 - Standard Test Methods for Rubber Properties in Tension.
D413 - Standard Test Method for Rubber Properties - Adhesion to Flexible Substrate.
D471 - Standard Test Method for Rubber Properties - Effect of Liquids.
D572 - Standard Test Method for Rubber - Deterioration by Heat and Oxygen.
D2240 - Standard Test Method for Rubber Properties - Durometer Hardness.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.1.1 Construction. The gaskets shall be molded synthetic rubber of the required dimensions and, unless otherwise specified (see 6.2), shall have six plies of cotton duck or synthetic woven fabric as the reinforcing material in the base (see figure 1).

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3.1.2 Reinforcing material. Unless otherwise specified (see 6.2), reinforcing material shall be synthetic woven fabric or cotton duck, firmly and evenly woven, and free from unsightly defects, knots, lumps, dirt, or irregularities of twist. The fabric shall be a basket or plain weave and weigh not less than 18 ounces per square yard. The warp and filler shall each have not less than 21 threads per linear inch. The fabric shall have a minimum tensile strength of 260 pounds both in the direction of the warp and the fill.

3.1.3 Rubber compound. The synthetic rubber compound when vulcanized shall conform to the requirements of tables I and II.

TABLE I. Physical requirements.

Tensile strength, minimum, psi	2,500
Tensile stress at 300 percent elongation, minimum, psi	1,200
Tensile strength after aging in oxygen bomb for 48 hours, percent of strength before aging, minimum	70
Ultimate elongation, minimum, percent	400
Ultimate elongation after aging in oxygen bomb for 48 hours, percent of elongation before aging, mininum	70
Compression set, maximum percent	15
Friction between rubber and fabric and between fabric plies, minimum, pounds per inch	15
Hardness, shore A, 3-second: minimum	55
maximum	65
Properties after immersion in test liquid:	
Tensile, strength, maximum change, percent	40
Ultimate elongation, maximum change, percent	30
Tensile stress at 300 percent elongation, maximum change, percent	30
Volume, maximum change, percent	30

TABLE II. Dimensional tolerances.

Tolerances in cross-sectional dimension:		
Flange thickness,	when 5/8 inch or less	±1/32 inch
	when over 5/8 inch	±1/16 inch
Overall height*,	when 2 inches or less	±1/16 inch
	when over 2 inches	±3/32 inch
Overall width,	when 6 inches or less	±1/16 inch
	when over 6 inches	±1/8 inch

* Overall height shall not differ by more than 1/32 inch when measured at any two points which are 2 inches apart along the length of the gasket.

3.1.4 Dimensions. The dimensions of the gasket, such as width, length, and height shall be as specified (see 6.2).

3.2 Identification marking. The rubber gasket shall be marked for identification in accordance with MIL-STD-190 (see 6.2).

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3.3 Workmanship. Workmanship shall be such that the product shall be free from imperfections which will adversely affect its general appearance, function, or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot size. For the purpose of sampling, a lot shall consist of all gaskets of the same size and length offered for delivery at one time.

4.3 Sampling.

4.3.1 Sampling for examination. Sampling for examination shall be in accordance with level II of MIL-STD-105.

4.3.2 Sampling for tests. Each lot of gaskets shall include the following for test purposes:

a. Twenty-four inch section of rubber gasket identical to the gasket submitted for the test.
b. One foot of unfrictioned synthetic woven fabric or cotton duck of the full width of the bolt or roll used in fabricating the gasket lot.

4.4 Examination. Each gasket shall be examined for compliance with the requirements specified in table III of this specification. The Acceptable Quality Level shall be 1.0 percent defective for major defects and 2.5 percent for minor defects, in accordance with MIL-STD-105. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

TABLE III. Classification of defects.

Category	Defect
Major:	
101	Length not as specified.
102	Surface not smooth.
103	Dimensions of width, height, and thickness not as specified.
Minor:	
201	Marking not as specified.

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4.5 Tests. The tests specified in table IV shall be conducted on the samples provided in 4.3.2 to determine conformance to 3.1.2 and 3.1.3. Any sample failing to pass the following tests, as applicable, shall be rejected and the lot represented by the sample shall be rejected. Test conditions shall be as specified in the applicable test method, unless otherwise specified (see 6.2).

TABLE IV. Test methods.

Test	ASTM Test Method	Fed. Std. 191 Test Method
Tensile strength	D412[1/]	
Tensile stress at 300 percent elongation	D412[1/]	
Ultimate elongation	D412[1/]	
Accelerated aging	D572	
Compression set	D395[2/]	
Adhesion test	D413[3/]	
Hardness	D2240[4/]	
Immersion test[5/]		
Tensile strength after immersion	D471	
Ultimate elongation	D471	
Tensile stress at 300 percent	D471	
Volume change	D471	
Thread count		5050
Tensile strength of fabric reinforcement		5100

1/ Tests shall conform to method A using straight specimens.

2/ Tests shall conform to method C.

3/ Tests shall conform to machine method.

4/ Tests shall be performed with type A durometer.

5/ Test fluid shall be water.

4.6 Packaging inspection. The preservation, palletization, packing, and marking of the gaskets shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation and packing. Each length of gasket shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage. Gaskets shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.2 Marking. In addition to any special marking required in the contract, each coil of gasket and each shipping container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Gaskets covered by this specification are for use on the seating surfaces of drydock caissons.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If reinforcing material is not required (see 3.1.1).
c. Reinforcing material, if different (see 3.1.2).
d. Dimensions and quantity of gaskets required (see 3.1.4).
e. Special marking, if required (see 3.2).
f. If testing conditions are different (see 4.5).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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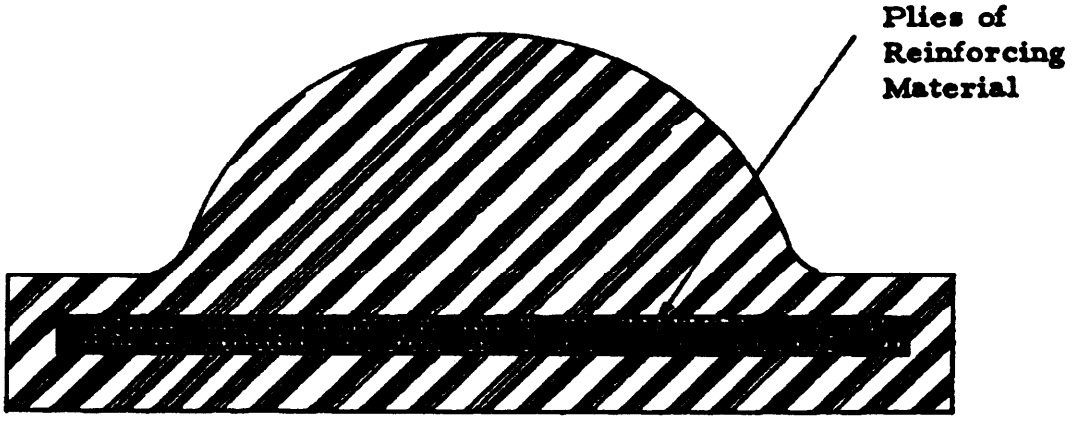


Figure 1. Cross-Section of Rubber Gasket

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